A 2-sectors, 2-factors general equilibrium model

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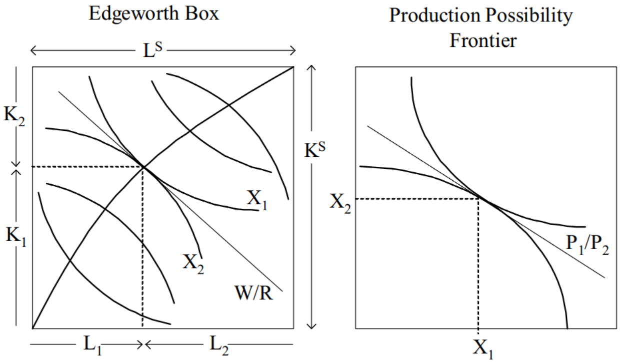
Outline

- Diagramatic exposition
- Mathematical exposition
- Deriving closed-form solution
- Calibration
- Simulations
- General representation
- 7-steps to become CGE modellers
- Gempack exercises

Diagramatic exposition



- Find K_1, L_1, K_2, L_2, the optimum allocation of factor endowment K^S, L^S.
- Endogenous variables: $K_1, L_1, K_2, L_2, X_1, X_2, P_1, P_2, W, R$
- Exogenous variables: K^S, L^S

Mathematically

- What we know already
  - TECHNOLOGY: Isoquant map of industry 1 and 2 $X_i = X_i(K_i, L_i)$
    - Suppose it is Cobb-Douglass $X_1 = K_1^{\alpha} L_1^{1-\alpha}$, $X_2 = K_2^{\beta} L_2^{1-\beta}$
  - PREFERENCE: Indifference curve of consumers $U = U(X_1, X_2)$
    - Suppose it is Cobb-Douglass $U = X_1^{\gamma} X_2^{1-\gamma}$
  - RESOURCE CONSTRAINT: Exogenous variables K_S, L_S.
- What we want to know
  - Quantity variables: the optimum K_1, L_1, K_2, L_2 and X_1, X_2
  - Price variables: the equilibrium P_1, P_2, W, R
- How? Draw Edgeworth Box and PPF/Indifference Curve (If you can)
- Solve it mathematically, assuming that
  - Firms is minimizing costs and household maximizing utility in a competitive market
  - This is the competitive solution $\approx$ Social planner solution (Invisible hand, 1st theorem of Welfare Economics)

Social planner's solution

A benevolent social planner maximizes $U = X_1^{\gamma} X_2^{1-\gamma}$ subject to $X_1 = K_1^{\alpha} L_1^{1-\alpha}$, $X_2 = K_2^{\beta} L_2^{1-\beta}$, $K_1 + K_2 = K_S$, $L_1 + L_2 = L_S$.

- Note that no prices here!

By substitution $X_2 = (K_S - K_1)^{\beta}(L_S - L_1)^{1-\beta}$, then

$$U = (K_1^{\alpha} L_1^{1-\alpha})^{\gamma}((K_S - K_1)^{\beta}(L_S - L_1)^{1-\beta})^{1-\gamma}, \text{ or}$$

$$U = K_1^{\alpha\gamma}(K_S - K_1)^{\beta(1-\gamma)} L_1^{(1-\alpha)\gamma}(L_S - L_1)^{(1-\beta)(1-\gamma)}$$

First order conditions of maximization:

$$\frac{\partial U}{\partial K_1} = 0 \text{ and } \frac{\partial U}{\partial L_1} = 0$$

Social planner's solution (cont.)

$$U = K_1^{\alpha\gamma}(K_S - K_1)^{\beta(1-\gamma)} L_1^{(1-\alpha)\gamma}(L_S - L_1)^{(1-\beta)(1-\gamma)}$$

$$\frac{\partial U}{\partial K_1} = 0 \Rightarrow \alpha\gamma K_1^{\alpha\gamma-1}(K_S - K_1)^{\beta(1-\gamma)} - K_1^{\alpha\gamma}\beta(1-\gamma)(K_S - K_1)^{\beta(1-\gamma)-1} = 0$$

$$\alpha\gamma K_1^{\alpha\gamma-1}(K_S - K_1)^{\beta(1-\gamma)} = K_1^{\alpha\gamma}\beta(1-\gamma)(K_S - K_1)^{\beta(1-\gamma)-1}$$

$$\alpha\gamma K_1^{-1} = \beta(1-\gamma)(K_S - K_1)^{-1} \Rightarrow \frac{\alpha\gamma}{K_1} = \frac{\beta(1-\gamma)}{K_S - K_1} \Rightarrow \frac{K_S - K_1}{K_1} = \frac{\beta(1-\gamma)}{\alpha\gamma}$$

$$\boxed{K_1 = \frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)}K_S} \text{ and } \boxed{K_2 = \frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)}K_S}$$

Social planner's solution (cont.)

$$U = K_1^{\alpha\gamma}(K_S - K_1)^{\beta(1-\gamma)} L_1^{(1-\alpha)\gamma}(L_S - L_1)^{(1-\beta)(1-\gamma)} \Rightarrow \frac{\partial U}{\partial L_1} = 0$$

$$(1-\alpha)\gamma L_1^{(1-\alpha)\gamma-1}(L_S - L_1)^{(1-\beta)(1-\gamma)}) - L_1^{(1-\alpha)\gamma}(1-\beta)(1-\gamma)(L_S - L_1)^{(1-\beta)(1-\gamma)-1} = 0$$

$$(1-\alpha)\gamma L_1^{-1} = (1-\beta)(1-\gamma)(L_S - L_1)^{-1} \Rightarrow \frac{(1-\alpha)\gamma}{L_1} = \frac{(1-\beta)(1-\gamma)}{L_S - L_1}$$

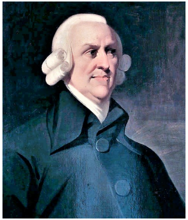
$$\Rightarrow \frac{L_S - L_1}{L_1} = \frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma}$$

$$\boxed{L_1 = \frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S} \text{ and } \boxed{L_2 = \frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S}$$

Social planner v.s. competitive market solution

"It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own self-interest. We address ourselves not to their humanity but to their self-love, and never talk to them of our own necessities, but of their advantages"

Adam Smith: The Wealth of Nations



Competitive solutions

- In a social optimum solution, a representative benevolent social planner (who know all the model's parameters), order the allocation of resource.
- In a competitive solution, firms and households don't care of others, maximizing profit and utility in a competitive market, responding to the price signals.
- Both solutions are pareto optimum $\rightarrow$ *The first fundamental theorem of welfare economics*
- Notes: We introduce prices, for both commodities and factors of production
  - P_1, P_2, W, R

Competitive market solution

- Producer's problem:
  - Producer 1: min $WL_1 + RK_1$ subject to $X_1 = K_1^{\alpha} L_1^{1-\alpha}$
  - Producer 2: min $WL_2 + RK_2$ subject to $X_2 = K_2^{\beta} L_2^{1-\beta}$
- Consumer's problem:
  - $\max U = X_1^{\gamma} X_2^{1-\gamma}$ subject to $Y = P_1 X_1 + P_2 X_2$
- Solution to producer's problem (see previous module):
  - Producer 1: $K_1 = \frac{\alpha c_1 X_1}{R}; L_1 = \frac{(1-\alpha) c_1 X_1}{W}$
  - Producer 2: $K_2 = \frac{\beta c_2 X_2}{R}; L_2 = \frac{(1-\beta) c_2 X_2}{W}$
  - Competitive market implies zero profit: $P_1 = c_1; P_2 = c_2$
- Solution to consumer's problem:
  - $X_1 = \gamma \frac{Y}{P_1}; X_2 = (1-\gamma) \frac{Y}{P_2} \Rightarrow \frac{P_1 X_1}{P_2 X_2} = \frac{\gamma}{1-\gamma} \Rightarrow P_2 X_2 = \frac{1-\gamma}{\gamma} P_1 X_1$

Competitive solution

- Producer 1: $K_1 = \frac{\alpha c_1 X_1}{R} \Rightarrow R = \frac{\alpha c_1 X_1}{K_1}$; $L_1 = \frac{(1-\alpha) c_1 X_1}{W} \Rightarrow W = \frac{(1-\alpha) c_1 X_1}{L_1}$
- Producer 2: $K_2 = \frac{\beta c_2 X_2}{R} \Rightarrow R = \frac{\beta c_2 X_2}{K_2}$; $L_2 = \frac{(1-\beta) c_2 X_2}{W} \Rightarrow W = \frac{(1-\beta) c_2 X_2}{L_2}$
- Equate R and use zero profit:
  $\frac{\alpha c_1 X_1}{K_1} = \frac{\beta c_2 X_2}{K_2} \Rightarrow \frac{\alpha P_1 X_1}{K_1} = \frac{\beta P_2 X_2}{K_2}$
- Substitute this: $P_2 X_2 = \frac{1-\gamma}{\gamma} P_1 X_1$

  $$\frac{\alpha P_1 X_1}{K_1} = \frac{\beta \frac{1-\gamma}{\gamma} P_1 X_1}{K_2} \Rightarrow \frac{\alpha}{K_1} = \frac{\beta(1-\gamma)}{\gamma K_2}$$
- Equate W and use zero profit:
  $\frac{(1-\alpha) c_1 X_1}{L_1} = \frac{(1-\beta) c_2 X_2}{L_2} \Rightarrow \frac{(1-\alpha) P_1 X_1}{L_1} = \frac{(1-\beta) P_2 X_2}{L_2}$
- Substitute this: $P_2 X_2 = \frac{1-\gamma}{\gamma} P_1 X_1$

  $$\frac{(1-\alpha) P_1 X_1}{L_1} = \frac{(1-\beta) \frac{1-\gamma}{\gamma} P_1 X_1}{L_2} \Rightarrow \frac{1-\alpha}{L_1} = \frac{(1-\beta)(1-\gamma)}{\gamma L_2}$$

Competitive solution

- We have found:

$$\frac{\alpha}{K_1} = \frac{\beta\,(1-\gamma)}{\gamma K_2} \text{ and } \frac{1-\alpha}{L_1} = \frac{(1-\beta)\,(1-\gamma)}{\gamma L_2}$$

- But we have resource constraint equations:

$$K^S = K_1 + K_2 \text{ and } L^S = L_1 + L_2$$

- Solve for K_1: $\frac{\alpha}{K_1} = \frac{\beta(1-\gamma)}{\gamma\left(K^S-K_1\right)} \Rightarrow \alpha\gamma K^S - \alpha\gamma K_1 = \beta\,(1-\gamma)\,K_1$
  $(\beta\,(1-\gamma) + \alpha\gamma)\,K_1 = \alpha\gamma K^S \Rightarrow K_1 = \frac{\alpha\gamma}{\beta(1-\gamma)+\alpha\gamma} K^S$
- Solve for L_1:
  $\frac{1-\alpha}{L_1} = \frac{(1-\beta)(1-\gamma)}{\gamma\left(L^S-L_1\right)} \Rightarrow (1-\alpha)\,\gamma L^S - (1-\alpha)\,\gamma L_1 = (1-\beta)\,(1-\gamma)\,L_1$
  $((1-\beta)\,(1-\gamma) + (1-\alpha)\,\gamma)\,L_1 = (1-\alpha)\,\gamma L^S$
  $\Rightarrow L_1 = \frac{(1-\alpha)\gamma}{(1-\beta)(1-\gamma)+(1-\alpha)\gamma} L^S$

Competitive solution: Factor allocation

The optimum for real (or quantity) of factor allocations

$$\boxed{K_1^* = \frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)} K_S} \quad \boxed{K_2^* = \frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)} K_S}$$

$$\boxed{L_1^* = \frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S} \quad \boxed{L_2^* = \frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S}$$

They depend on (1) factor endowment, (2) technology, and (3) preference *And they are exactly the same as the social-planner solution!*

Competitive solution: Output

Substitute optimal factor allocation into production function:

$$X_1 = K_1^{\alpha} L_1^{1-\alpha}, \; X_2 = K_2^{\beta} L_2^{1-\beta}$$

$$\boxed{X_1^* = \left(\frac{\alpha\gamma}{\alpha\gamma + \beta(1-\gamma)} K_S\right)^{\alpha} \left(\frac{(1-\alpha)\gamma}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S\right)^{1-\alpha}}$$

$$\boxed{X_2^* = \left(\frac{\beta(1-\gamma)}{\alpha\gamma + \beta(1-\gamma)} K_S\right)^{\beta} \left(\frac{(1-\beta)(1-\gamma)}{(1-\alpha)\gamma + (1-\beta)(1-\gamma)} L_S\right)^{1-\beta}}$$

Competitive solution: Prices

Relative factor prices W/R depends on the factor intensity K_1/L_1 or K_2/L_2 (Homogeneity in production function)

$$\frac{W}{R} = \frac{1-\alpha}{\alpha}\frac{K_1^*}{L_1^*} = \frac{1-\alpha}{\alpha}\frac{\frac{\alpha\gamma}{\alpha\gamma+\beta(1-\gamma)}K_S}{\frac{(1-\alpha)\gamma}{(1-\alpha)\gamma+(1-\beta)(1-\gamma)}L_S}$$

Relative commodity prices P_1/P_2 depends on rasio of optimum consumption X_2/X_1 (Homogenity in utility function)

$$\frac{P_1}{P_2} = \frac{\gamma}{1-\gamma}\frac{X_1^*}{X_2^*}$$

Competitive solution: Prices

GE model is basically a "corn" economy. We can use one commodity/factor as a *numeraire*. Suppose P_2 is that *numeraire*. Then all prices are relative to P_2. The competitive solutions of prices become:

$$P_2 = 1 \quad P_1 = \frac{\gamma}{1-\gamma}\frac{X_1^*}{X_2^*}$$

$$W = \frac{(1-\beta)\,X_2^*}{L_2^*} \quad R = \frac{\beta X_2^*}{K_2^*}$$

Competitive solution: Walras Law

- Walras' law is a principle in general equilibrium theory asserting that budget constraints imply that the values of excess demand (or, conversely, excess market supplies) must sum to zero.
- It imples that when there is N market in the economy and the $N-1$ market is in equilibrium, then the $N-th$ market must be in equalibrium.
- Nominal homogenity $\Rightarrow$ if all prices double, no change in quantities or real variables
- Only relative prices matters.

Empirical GE model: Calibration

- Data (of the economy) represent initial equilibrium: *Important Assumption!!*
- What kind of data? For example: Input-Output Table

	Industry 1	Industry 2	Consumption	Total
1			P_1X_1	P_1X_1
2			P_2X_2	P_2X_2
Salary	WL_1	WL_2		WL^S
Surplus	RK_1	RK_2		RK^S
Total	P_1X_1	P_2X_2		Y

Calibration

- I-O Table Indonesia (2000)

	Industry 1	Industry 2	Consumption	Total
1			347	347
2			953	953
Salary	68	340		408
Surplus	279	613		892
Total	347	953	1300	1300

- CGE model has to represent the data.
- If solved, it has to be able to reproduce this data (initial equilibrium).
- Calibration means "finding parameters of GE model (i.e. α, β, γ) that can reproduce the initial equilibrium exactly like this data".

Calibration

- From Input-Output Table, $WL_1 = 68.05$, $WL_2 = 340.13$, $RK_1 = 278.55$, and $RK_2 = 613.17$.
- Then, $P_1X_1 = WL_1 + RK_1 = 346.6$, and $P_2X_2 = WL_2 + RK_2 = 953.3$.
- Then, derived from first order conditions

$$
\begin{aligned}
\alpha &= \frac{RK_1}{P_1X_1} = 0.804 \\
\beta &= \frac{RK_2}{P_2X_2} = 0.643 \\
\gamma &= \frac{P_1X_1}{P_1X_1 + P_2X_2} = 0.267
\end{aligned}
$$

- Now, we have all the parameters!

Interpretation of GE initial solutions

- Interpretation of prices. $P_1 = 1, P_2 = 1.17, W = 0.61, R = 0.61$. Answer the following questions?
  - Can we find prices in Rupiahs? Why?
  - Can we say $P_2 > P_1$ or $P_1 > W$? Why?
  - Can we say $W = R$? Why?
- Interpretation of quantity.
  $K_1 = 278, K_2 = 613, L_1 = 68.05, L_2 = 340.13, X_1 = 211.22, X_2 = 497$.
  - What is the unit of labor? Man-Hour?
  - What is the unit of K?
  - Can we say $X_1 < X_2$? Why?
  - Can we say $K_1 < K_2$ or $L_1 < L_2$?
- The value of CGE initial solutions depend on what is the numeraire (for prices), and the initial value of W and R that we "guess" (for quantities).

CGE and comparative static

- If those solutions tell nothing, what is CGE for???
- Here is the biggest magic: *CGE model will tell you everything in comparative static analysis!*
- Comparative static: change the exogenous variables, K^S, L^S (simulation), solve again, and compare it with initial solution.
- The (percentage) change of all the variables are *independent* on initial value we "guess", and are comparable among them

Effect of change in factor endowment

See illustration in Excel file (ge.xlsx).

More general representation of CGE model

Firm i's problem, where $i \in 1, 2$

$$\min_{K_i, L_i} W \cdot L_i + R \cdot K_i \quad \text{s.t.} \quad X_i = X_i\left(K_i, L_i\right)$$

and this will produce demand for factors

$$K_i = K_i\left(W, R, X_i\right) \text{ and } L_i = L_i\left(W, R, X_i\right)$$

Household's problem is

$$\max_{Q_i} U\left(Q_i\right) \text{ s.t } \sum P_i Q_i = Y \text{s.t.} Y = W \cdot L^S + R \cdot K^S$$

and this will produce demand for commodity

$$Q_i = Q_i\left(P_i, Y\right)$$

With previous example: CD Technology/Preferences

1. $K_i = \frac{\alpha_i c_i X_i}{R} \Leftarrow$ Demand for capital
2. $L_i = \frac{(1-\alpha_i) c_i X_i}{W} \Leftarrow$ Demand for labor
3. $c_i = \left(\frac{R}{\alpha_i}\right)^{\alpha_i} \left(\frac{W}{1-\alpha_i}\right)^{1-\alpha_i} \Leftarrow$ Unit cost definition
4. $P_i = c_i$, or $P_i X_i = R \cdot K_i + W \cdot L_i \Leftarrow$ zero profit
5. $Q_i = \gamma_i \frac{Y}{P_i} \Leftarrow$ Demand for commodities
6. $Y = W \cdot L^S + R \cdot K^S \Leftarrow$ Household income
7. $K^S = \sum_i K_i \Leftarrow$ Market clearing for capital
8. $L^S = \sum_i L_i \Leftarrow$ Market clearing for Labor
9. $Q_i = X_i \Leftarrow$ Market clearing for commodities

Counting variables and equations

- From equation 1 to 9, and $i = 1, 2$, we have
  - 15 equations, to determine
  - 17 variables: $K_1, L_1, X_1, K_2, L_2, X_2, Q_1, Q_2, c_1, c_2, P_1, P_2, W, R, Y, K^S, L^S$
- To close the system, by Walras' law one market clearing will be dropped. It gives now:
  - 14 equations, to determine
  - 14 variables, because K^S, L^S are exogenous, and one of the price, let's say P_1 will be used as numeraire.

With CES technology and CD preference

1. $K_i = X_i \alpha_i^{\sigma_i} \left(\frac{R}{c_i}\right)^{-\sigma_i} \Leftarrow$ Demand for capital
2. $L_i = X_i \beta_i^{\sigma_i} \left(\frac{W}{c_i}\right)^{-\sigma_i} \Leftarrow$ Demand for labor
3. $c_i = \left(\alpha_i^{\sigma_i} R^{1-\sigma_i} + \beta_i^{\sigma_i} W^{1-\sigma_i}\right)^{\frac{1}{1-\sigma_i}} \Leftarrow$ Unit cost definition
4. $P_i = c_i$, or $P_i X_i = R \cdot K_i + W \cdot L_i \Leftarrow$ zero profit
5. $Q_i = \gamma_i \frac{Y}{P_i} \Leftarrow$ Demand for commodities
6. $Y = W \cdot L^S + R \cdot K^S \Leftarrow$ Household income
7. $K^S = \sum_i K_i \Leftarrow$ Market clearing for capital
8. $L^S = \sum_i L_i \Leftarrow$ Market clearing for Labor
9. $Q_i = X_i \Leftarrow$ Market clearing for commodities

7 Steps to become a CGE modeler with Gempack

1. Understand the theory (in equations)
2. Linearization
3. Get and prepare the data
4. Creating Tablo file
5. Creating database file
6. Creating simulation file
7. Simulation

Step 1 - The theory

1. $K_i = X_i \alpha_i^{\sigma_i} \left(\frac{R}{c_i}\right)^{-\sigma_i} \Leftarrow$ Demand for capital
2. $L_i = X_i \beta_i^{\sigma_i} \left(\frac{W}{c_i}\right)^{-\sigma_i} \Leftarrow$ Demand for labor
3. $c_i = \left(\alpha_i^{\sigma_i} R^{1-\sigma_i} + \beta_i^{\sigma_i} W^{1-\sigma_i}\right)^{\frac{1}{1-\sigma_i}} \Leftarrow$ Unit cost definition
4. $P_i = c_i$, or $P_i X_i = R \cdot K_i + W \cdot L_i \Leftarrow$ zero profit
5. $Q_i = \gamma_i \frac{Y}{P_i} \Leftarrow$ Demand for commodities
6. $Y = W \cdot L^S + R \cdot K^S \Leftarrow$ Household income
7. $L^S = \sum_i L_i \Leftarrow$ Market clearing for Labor
8. $Q_i = X_i \Leftarrow$ Market clearing for commodities

Why no market clearing for capital?

Step 2. Linearization (percentage change)

1. $k_i = x_i - \sigma_i (r - c_i) \Leftarrow$ Demand for capital
2. $l_i = x_i - \sigma_i (w - c_i) \Leftarrow$ Demand for labor
3. $c_i = S_i^L \cdot w + \left(1 - S_i^L\right) \cdot r \Leftarrow$ Unit cost definition
4. $P_i X_i \cdot (p_i + x_i) = WL_i \cdot (w + l_i) + RK_i (r + k_i) \Leftarrow$ zero profit
5. $q_i = y - p_i \Leftarrow$ Demand for commodities
6. $Y \cdot y = WL^S \cdot \left(w + l^S\right) + RK^S \cdot \left(r + k^S\right) \Leftarrow$ Household income
7. $x_i = q_i \Leftarrow$ Market clearing for commodities
8. $WL^S \cdot l^S = \sum_i WL_i \cdot l_i \Leftarrow$ Market clearing for Labor

Step 3. Get and prepare the data

Input-Output table:

	Industry 1	Industry 2	Consumption	Total
1			347	347
2			953	953
Salary	68	340		408
Surplus	279	613		892
Total	347	953	1300	1300

- WL_i and RK_i are base coefficient.
- Other coefficient, $P_iX_i = WL_i + RK_i, WL^S = \sum_i WL_i, RK^S = \sum_i RK_i$, $Y = WL^S + RK^S$.

Step 4. Creating Tablo File

File structure

1. FILE statement. Declaring a 'logical' file name where your data will be located.
2. SET statement. Declaring set names, or subscript in the model.
3. VARIABLE statement. Declaring all the variables in the model.
4. COEFFICIENT statement. Declaring all the coefficient and parameter in the model.
5. READ statement. Asking the program to read the 'base' coefficient from database.
6. FORMULA statement. Calculation of other coefficient as a function of base coefficient.
7. UPDATE statement. Updating the base coefficient.
8. EQUATION statement. Listing all the model equations.

Step 5. Creating Database File

- Database file record the 'base' coefficient, here the coefficients are WLi and RK_i.
- In GEMPACK the database file has the extension .har, and created and read by a GEMPAK program called VIEWHAR.

Step 6. Creating Simulation File

- Simulation file is a text file with extension .CMF where you put the closure and shock statement, and some other ancilliary information.
- *Closure* is a statement declaring which variables are endogenous, and which variables are exogenous that make sure that the number of endogenous variables is equal to the number of statement and also will tell us what variables that can be shocked.
- *Shock statement* is changing one or combination of exogenous variables (our policy question) to see its impact on all the endogenous variables.

Step 7. Simulation

Asking GEMPACK to implement our 6 steps. Now, it is your first time for computer exercise with Gempack!!